ModelIt!

Student Activity Pack

The Disappearing Act

Ice Melts, But the Mass Never Leaves

Grade 5 | NGSS 5-PS1-2 | Scale, Proportion, and Quantity

STUDENT NAME _____ DATE _____

ModelIt! — Discovery Collective

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 An ice cube weighs 50 grams. It melts completely. How much does the water weigh?

2 When wood burns, the ash weighs less than the wood. Where did the mass go?

3 Draw what happens to atoms inside an ice cube when it melts.

4 Does matter disappear or change form? Explain your thinking.

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Air Temperature | Starting Mass | Ice Amount | Liquid Water Amount

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Room Temperature

Normal temperature conditions

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Freezing

Lock Temperature to 10%

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Melting

Lock Temperature to 90%

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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> Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Water Vapor Amount — Gas state when water evaporates

Time — Duration of heating or cooling

Container Type — Open vs. closed system

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

How does the total mass of matter stay the same when it changes form? Use evidence from your simulation to support your answers.

1. **Does it matter if matter is solid or liquid - is the mass the same?**

2. **Why does ice LOOK like it's disappearing when it melts?**

3. **Can you think of other examples where matter changes form but not mass?**

4. **If you weigh yourself before and after breathing out, are you lighter?**

5. **When you eat food, does it disappear? Where does the mass go?**

8 STEM Challenge: Solve the Burning Mystery

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YOUR ENGINEERING MISSION

Explain where the 'missing' mass goes when wood burns and only ash remains.

THE SCENARIO

A 10 kg log burns down to 0.5 kg of ash. Where did 9.5 kg of mass go?

Engineering Design Thinking

1. WHAT WOULD YOU NEED TO CAPTURE TO PROVE CONSERVATION?
2. WHAT EQUIPMENT WOULD YOU NEED?
3. HOW WOULD YOU WEIGH GASES?
4. DRAW YOUR EXPERIMENTAL SETUP.

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Forensic Chemists use conservation of matter to solve crimes - they can reconstruct what substances were present at crime scenes. They earn $65,000-$100,000/year.